READER'S GUIDE · TOOL 09

How to read the *air calculator.*

The calculator takes one number, an AQI, and answers three plain questions: **what is in the air, how much of it do I breathe in, and what should I do?** This guide explains every tab and every number, the maths behind them, what was assumed and where the figures come from.

WISEWATT · WISEWATT.IN/AIR-CALCULATOR.HTML · INDIA CPCB SCALE · EDUCATIONAL, NOT MEDICAL ADVICE

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01 · THE IDEA IN ONE MINUTE

An AQI is a grade, *not a weight.*

AQI 350 tells you the air is in the "Very Poor" band. It does not tell you how much pollution you actually take into your body. That depends on three things: **how dirty the air is, how hard you are breathing and for how long.**

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Inhaled PM2.5  =  Concentration  ×  Breathing rate  ×  Time
     (µg)            (µg per m³)      (m³ per hour)     (hours)
```

One worked example runs through this guide. Enter **AQI 350** and the calculator says:

STEP	RESULT	IN PLAIN WORDS
AQI → PM2.5	185 µg/m³	Each cubic metre of air holds about 185 micrograms of fine particles, if PM2.5 is what set the AQI.
Sitting for 8 hours	373 µg	A resting adult breathes about 2 m³ of air in that time.
Running for 1 hour	555 µg	One hour of running takes in more than eight hours of sitting.
Against WHO	12.3×	About twelve times the WHO 24-hour reference level of 15 µg/m³.

02 · THREE NUMBERS TO KNOW

What each kind of number *means.*

NUMBER	UNIT	WHAT IT TELLS YOU	WHAT IT DOES NOT TELL YOU
AQI	0 to 500, no unit	How bad the worst pollutant is today, as a grade in six bands.	Which pollutant it is, or any weight of pollution.
PM2.5 concentration	µg/m³	The weight of fine particles in one cubic metre of air.	How much you personally inhale.
Inhaled mass	µg or mg	The weight of fine particles that entered your nose and mouth over a period.	How much stayed in your lungs, or any measure of harm.

The six CPCB bands, with the colours used in the calculator. Every colour is always shown with its name.

BAND	AQI	PM2.5 (MG/M³)	CPCB HEALTH STATEMENT, IN SHORT
Good	0–50	0–30	Minimal impact.
Satisfactory	51–100	31–60	Minor breathing discomfort for sensitive people.
Moderate	101–200	61–90	Discomfort for people with lung, asthma or heart conditions.
Poor	201–300	91–120	Discomfort for most people on prolonged exposure.
Very Poor	301–400	121–250	Respiratory illness on prolonged exposure.
Severe	401–500	above 250	Affects healthy people; serious impact on those with existing disease.

03 · TAB BY TAB

What each tab *shows you.*

All ten tabs share the same inputs. Change the AQI, your age group or your activity in one place and every tab updates.

TAB 1 — HOME

The answer at a glance

YOU ENTER	An AQI between 0 and 500.
YOU SEE	The band, the PM2.5 estimate in µg and mg, the WHO multiple, the inhaled mass for your chosen activity, and what to do at this level.
HOW TO READ IT	The dial shows where your number sits on the six bands. **Copy report** gives a text summary you can paste into a message.

TAB 2 — AQI CONVERTER

From an index back to a concentration

YOU ENTER	An overall AQI. Or, lower down, your own readings for up to eight pollutants.
YOU SEE	The PM2.5 estimate, the formula with your numbers filled in, and the full CPCB breakpoint table.
HOW TO READ IT	The PM2.5 figure is **conditional**. It is right if PM2.5 is the pollutant that set the AQI. If PM10 or a gas set it, real PM2.5 is lower. So treat it as an upper estimate.

TAB 3 — PM2.5 CALCULATOR

Use a real reading if you have one

YOU ENTER	A measured PM2.5 value in µg/m³, from a monitor or an official station.
YOU SEE	Its CPCB sub-index and band, the same value in mg/m³, and a one-cubic-metre box of air with that load in it. You can also compare two places.
HOW TO READ IT	Once entered, the whole calculator uses your measured value in place of the AQI estimate. 1 mg = 1,000 µg, so 185 µg/m³ is 0.185 mg/m³.

TAB 4 — HUMAN EXPOSURE

How much you breathe in

YOU ENTER	Age group, activity and duration. Advanced users can type their own breathing rate.
YOU SEE	Air volume breathed, PM2.5 inhaled in µg and mg, the calculation line by line, and a "what if" table.
HOW TO READ IT	This is the mass entering the nose and mouth. Part of it is breathed out again. It is useful for **comparing choices**, such as running outdoors against resting indoors.

TAB 5 — ACTIVITY SIMULATOR

Same air, different effort

YOU ENTER	Age group and duration.
YOU SEE	A bar chart and table of inhaled mass for nine activities, from sleeping to heavy labour.
HOW TO READ IT	For an adult, running moves about **twelve times** more air than sitting. Effort matters as much as the AQI.

TAB 6 — INDOOR VS OUTDOOR

Your whole day, block by block

YOU ENTER	Blocks of your day: place (indoors or outdoors), effort and hours. Indoor PM2.5 if you have measured it, or an assumed indoor/outdoor ratio.
YOU SEE	The day's total, the split between indoor and outdoor hours, and a line that climbs through the day.
HOW TO READ IT	The example day at AQI 350 totals about **1,321 µg**: 852 from 21 indoor hours and 468 from 3 outdoor hours. Three hours outside give over a third of the day's load.

TAB 7 — WHO COMPARISON

A reference line, not a verdict

YOU ENTER	Nothing new.
YOU SEE	Your PM2.5 beside the WHO 24-hour level (15 µg/m³) and, separately, the WHO annual level (5 µg/m³), with India's standards for context.
HOW TO READ IT	The 24-hour level applies to a full-day average. The annual level applies only if air like this lasted a year. WHO found no level below which PM2.5 is harmless, so lower is always better.

TAB 8 — HEALTH BY AGE

Precautions for you

YOU ENTER	Age group and any of: asthma, COPD, heart disease, diabetes, pregnancy, other lung condition.
YOU SEE	General advice for the band, a note for your age group, and extra precautions for each condition you selected. Below that, how pollution affects lungs, heart, brain, children and pregnancy.
HOW TO READ IT	This is education, not diagnosis. There are no risk percentages here, on purpose.

TAB 9 — PROTECTION SIMULATOR

Masks and purifiers, as "what if"

YOU ENTER	Whether you wear a well-fitted N95 outdoors and run a purifier indoors, and how much reduction you assume for each.
YOU SEE	Your day with and without protection, split by outdoor and indoor hours.
HOW TO READ IT	With an assumed 70% respirator reduction outdoors, the example day falls from 1,321 to about **993 µg**. The day falls by less than 70% because the mask covers only three hours.

TAB 10 LEARN THE SCIENCE

The background

YOU SEE

How small PM2.5 is next to a human hair, where particles land in the body, how India's AQI is built, why long-term exposure matters, and the full source list.

04 · THE MATHS IN ONE PLACE

Five formulas, *nothing hidden.*

1. AQI to PM2.5

CPCB draws a straight line between breakpoints. The calculator runs that line backwards.

```
C = B_low + (AQI - I_low) ÷ (I_high - I_low) × (B_high - B_low)

AQI 350 sits in the 301–400 band, where PM2.5 runs from 120 to 250:
C = 120 + (350 - 300) ÷ 100 × 130
C = 185 µg/m³
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2. Micrograms to milligrams

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mg = µg ÷ 1,000
185 µg/m³ = 0.185 mg/m³
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3. Inhaled mass

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D = C × IR × T

Adult sitting for 8 hours at 185 µg/m³:
D = 185 × 0.252 × 8
D = 373 µg = 0.373 mg
```

4. A whole day

Total = D_1 + D_2 + D_3 + …

Each block of the day has its own C, IR and T.

Indoor C = Outdoor C × indoor/outdoor ratio

185 × 0.6 = 111 µg/m³

5. Protection and the WHO multiple

Adjusted C = C × (1 - assumed reduction)

With an assumed 70% reduction while the mask is on:

185 × (1 - 0.70) = 55.5 µg/m³

WHO multiple = C ÷ guideline

185 ÷ 15 = 12.3

Breathing rates used (m³ of air per hour)

Mean values from the US EPA Exposure Factors Handbook, Table 6-2, converted from m³ per minute.

AGE GROUP	SLEEP	SEDENTARY	LIGHT	MODERATE	HIGH
0–5 years	0.26	0.27	0.66	1.26	2.22
6–12 years	0.27	0.29	0.66	1.32	2.52
13–18 years	0.30	0.32	0.78	1.50	2.94
19–35 years	0.26	0.25	0.72	1.56	3.00
36–60 years	0.30	0.29	0.78	1.68	3.12
60+ years	0.31	0.29	0.72	1.56	2.82

EFFORT LEVEL	ACTIVITIES IN THE CALCULATOR
Sleep	Sleeping
Sedentary	Sitting, resting
Light	Normal walking, light exercise
Moderate	Brisk walking, cycling, moderate exercise
High	Running, heavy physical labour

05 · ASSUMPTIONS

What the calculator *takes for granted.*

- **PM2.5 set the AQI.** When you enter only an AQI, the PM2.5 figure assumes PM2.5 was the worst pollutant. Otherwise the true value is lower.
- **The AQI is a 24-hour figure.** CPCB's index uses 24-hour averages for particles. A reading at one moment can be higher or lower.
- **Above AQI 400, the line continues.** CPCB gives no upper PM2.5 value for "Severe". The calculator extends the line to 380 µg/m³ at AQI 500 and marks these values as indicative.
- **Breathing rates are population averages** from a US dataset, matched to the nearest age band. Your own rate depends on body size, fitness and effort.
- **Activities share five effort levels.** Brisk walking and cycling, for example, use the same rate.
- **Indoor air is a ratio of outdoor air** unless you enter a measurement. The default is 0.6 with windows closed and 0.9 with them open. Cooking, incense or smoking can push indoor air above outdoor air.
- **Mask and purifier reductions are yours to set.** The starting values (70% and 50%) are illustrations, not measured performance.
- **The mask is worn only outdoors and the purifier runs only indoors.**

06 · PRECAUTIONS BUILT IN

What the calculator deliberately *does not do.*

It will not invent harm numbers

- No percentage of lung damage.
- No years of life lost.
- No probability of disease or death.
- No age-based "damage multiplier".

It labels what is uncertain

- Estimates from AQI are marked "conditional".
- Ratios and reductions are marked "assumption".
- WHO levels are shown as a reference, with their averaging period.
- Health text describes population research, not your diagnosis.

Why no harm score? Inhaled mass can be calculated. The share that stays in the lungs depends on particle size and how each person breathes, so it is left out. Disease risk comes from long-term studies of large populations and cannot be worked out for one person from one AQI reading.

07 · LIMITS

Where to be *careful.*

- This is an educational tool. It does not replace advice from your doctor.
- PM2.5 is the only pollutant carried through the exposure tabs. Ozone, nitrogen dioxide and others matter too.
- Adding up micrograms over days is a fair way to compare routines. It is not a measure of damage stored in the body.
- The calculator does not fetch live data. You bring the AQI or the PM2.5 reading.
- Use the results to compare options and to decide when to cut back outdoor effort, not as precise personal doses.

08 · SOURCES AND GLOSSARY

Where the numbers *come from.*

SOURCE	USED FOR
Central Pollution Control Board, National Air Quality Index	Bands, breakpoints for all eight pollutants, sub-index rule, health statements.
WHO Global Air Quality Guidelines, 2021	Guideline levels of 15 µg/m³ (24-hour) and 5 µg/m³ (annual), interim targets.
US EPA Exposure Factors Handbook, Chapter 6, 2011	Breathing rates by age and effort level (Table 6-2).
WHO fact sheet: ambient air quality and health	Summary of health effects.
India National Ambient Air Quality Standards, 2009	PM2.5 standards of 60 µg/m³ (24-hour) and 40 µg/m³ (annual), shown for context.

Glossary

AQI	Air Quality Index. A 0–500 grade built from the worst pollutant of the day.
CPCB	Central Pollution Control Board, the agency that defines India's AQI.
PM2.5	Particles 2.5 micrometres across or smaller, about 30 times thinner than a hair. Small enough to reach the deep lung.
PM10	Particles up to 10 micrometres. Includes PM2.5.
µg/m³	Micrograms per cubic metre of air. A microgram is one millionth of a gram.
Sub-index	The 0–500 score for one pollutant. The highest sub-index becomes the AQI.
Breathing rate	Volume of air breathed per hour, in cubic metres. Also called inhalation rate.
24-hour average	The mean concentration over a full day. Both the AQI and the WHO short-term level use it.
N95	A respirator that filters at least 95% of test particles in a laboratory. Real protection depends on fit.